Notes of Relative Szemerédi's Theorem

Notation

- $[N]$: $\{1, 2, \cdots, N\}$
- $[a, b]$: $\{a, a+1, \cdots, b\}$ for $a < b$

Szemerédi's Theorem : Natural Version (NAT-SZEM)

Let N be a positive integer and $Z_N := \mathbb{Z}/N\mathbb{Z}$. Let $\delta > 0$ be a real number and $k \geq 3$ be an integer. There exists a positive integer $N_0(\delta, k)$ such that for $N > N_0(\delta, k)$ with $(N, (k-1)!) = 1$ if $A \subset Z_N$ with $|A| \geq \delta N$ then it satisfies that A contains a non-trivial arithmetic progression of lenght k.

Practically N will be a prime (hence $(N, (k-1)!) = 1$ and large enough to capture at-least some constant multiplicative factor (depending on δ of N^2 (as a result an equivalent counting version as stated below) and in particular should be $\gtrsim_\delta \lfloor N^2/k \rfloor + O(k+1)$ considering the minimal starting point of the final k-arithmetic progression. We state the following counting version of Szemeredi's Theorem whose equivalence to the aforementioned theorem was given by Vernavides' in 1959 using an averaging argument

Szemerédi's Theorem : Counting Version (COUNT-SZEM)

Given a positive δ and an integer $k \geq 3$ with the conditions from NAT-SZEM if $A \subset Z_N$ then A contains $\gtrsim_\delta N^2$ non-trivial and unique arithmetic progressions.

Also note that since there exists a (Freiman) map from Z_N to $[2N-1]$ (to prevent wrap arounds) which preserves arithmetic progressions (and all additive relations with at most 4 elements) in a natural manner it suffices to interchangeably use $[N]$ and Z_N (where these two N are not necessarily the same). With this agreed the aforementioned version of the theorem states that given any subset of $[N]$, with N being sufficiently large, such that $\limsup_{N\to\infty} \frac{|A\cap[N]|}{N} > \delta > 0$ (or A having positive (upper) density) contains many (at-least a multiplicative factor of N^2) arithmetic progressions. The expression concerning the limit loosely says that if as N gets bigger then if more number of elements of $[N]$ are inside $A \cap [N]$ than outside, then A is large enough to contain many arithmetic progressions.

In the spirit of the proof of relative Szemeredi's theorem (stated later) given by Conlon, Fox and Zhao, a weighted version of Szemeredi's theorem (a crucial portion in the proof) is stated below. The equivalence of these theorems are shown below the statement.

Szemerédi's Theorem : Weighted Version (FUNC-SZEM)

Given any real number $\delta > 0$ and an integer $k \geq 3$ there exists a positive constant c=c(δ, k) such that for a measure $f : Z_N \to [0,1]$ and $\mathbb{E}f \geq \delta$ with N sufficiently large (in previously stated sense) satisfies the following

$$\mathbb{E}_{x,d\in Z_N}[f(x)f(x+d)\cdots f(x+(k-1)d)] \geq c - o_{\delta,k}(1)$$

The second term on the right side denotes a quantity which goes to zero as N (which depends on δ and k) grows larger.

We show the equivalence of the weighted version and the natural version.

$\boxed{\text{FUNC-SZEM} \implies \text{NAT-SZEM}}$

Pick $A \subset Z_N$ such that $|A| \geq \delta N$. Set $f := \mathbf{1}_A$ to be the characteristic function of A. Next *a priori* $\mathbb{E}f \geq \delta$

$$\begin{aligned} & \mathbb{E}_{x,d \in Z_N}[f(x)f(x+d)\cdots f(x+(k-1)d)] \geq c - o_{\delta,k}(1) \\ \implies & \mathbb{E}_{x,d \in Z_N}[\mathbf{1}_A(x)\mathbf{1}_A(x+d)\cdots \mathbf{1}_A(x+(k-1)d)] \gtrsim_{\delta,k} 1 \end{aligned}$$

The last quantity implies that the number of k-APs in A $\gtrsim_\delta N^2$. So in particular it implies the existence of many k-APs in A and the natural version to be specific.

$\boxed{\text{NAT-SZEM} \implies \text{FUNC-SZEM}}$

Since the natural version implies the counting version, set $A := \{x \in Z_N : f(x) \geq \delta/2\}$. Since $\mathbb{E}f \geq \delta$ it follows definitionally,

$$\frac{1}{N}\sum_{x \in Z_N} f(x) \geq \delta \implies \frac{1}{N}\sum_{x \in A} f(x) + \frac{1}{N}\sum_{x \notin A} f(x) \geq \delta$$

Since $0 \leq f \leq 1$, the first part is $|A|/N$. By definition, the second part is bounded above by $\delta/2$. This finally implies $|A| \geq \delta N/2$. Next it follows that,

$$\begin{aligned} & \mathbb{E}_{x,d \in Z_N}[f(x)f(x+d)\cdots f(x+(k-1)d)] \\ & \geq \mathbb{E}_{x,d \in Z_N}[\mathbf{1}_A(x)\mathbf{1}_A(x+d)\cdots \mathbf{1}_A(x+(k-1)d)] \gtrsim_{\delta,k} 1 \end{aligned}$$

The last inequality follows by restricting $f \in [\delta/2, 1]$ and then using the counting version. Alternatively, the trivial inequality $f \geq f \cdot 1_A$ results the same.

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As a theorem this is concerned with finding arithmetic progressions in a sufficiently dense subset of Z_N. An equivalent version of Szemeredi's theorem may be stated as the existence of arithmetic progressions in a relatively dense subset of the integers which can be the dense analog of the A. This is because, naturally the integers are wrapped around module N and hence any dense subset of A of $\mathbb{Z}/N\mathbb{Z}$ can be unwrapped in the natural sense to yield a dense subset of the integers. To be precise $\widetilde{A} := \bigcup_{d\in\mathbb{Z}_{\geq 0}}(A+k\cdot d) \subset \bigcup_{d\in\mathbb{Z}_{\geq 0}} \mathbb{Z}_{N+k\cdot d} = \mathbb{Z}$ and *a priori* the specified union will be dense in integers. This can be pictorially viewed by the commutative diagram (in two parts) as below:

$$\begin{tikzcd}
A \arrow[r, hook, "i_A"] \arrow[dr, hook, "(\bmod N)^{-1}"'] & \mathbb{Z}/(N) \arrow[d, "(\bmod N)^{-1}"] \\
& \mathbb{Z}
\end{tikzcd}
\qquad
\begin{tikzcd}
A \arrow[d, "(\bmod N)^{-1}"'] \arrow[dr, hook, "(\bmod N)^{-1}"] & \\
\widetilde{A} \arrow[r, hook, "i_{\widetilde{A}}"'] & \mathbb{Z}
\end{tikzcd}$$

Alternatively if $\widetilde{A} \subset \mathbb{Z}$ is dense and contains an arithmetic progression of length k, then these arithmetic progressions may be naturally embedded in Z_N for some $N = N(k)$ chosen by padding the initial part with integers till the start of the AP adjoined with the rest of the AP and the gaps in between two consecutive integers filled in the natural order.

Szemeredi's theorem is deep and it had major developments in proof involving combinatorial methods (as in the original proof given by Szemeredi) to ergodic theory (as a breakthrough by Furstenberg). Another important proof uses a hypergraph regularity lemma (a weaker version of which will be used to prove the 3-AP case of Szemeredi's theorem). Here this theorem will be assumed as a black box and used as the final element to prove relative Szemeredi's theorem. Since the complexity of extending the proof of Roth's theorem (3-AP case of Szemeredi) to a proof for Szemeredi's theorem is notational, for simplicity purposes relative Roth's theorem will be proved. We state and prove a contrapositive version of Roth's theorem as follows

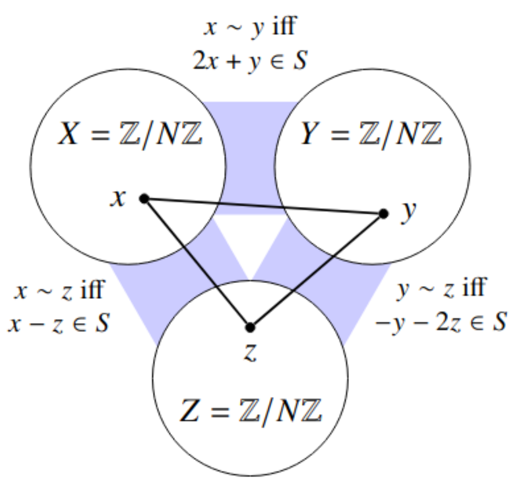
Roth's Theorem

Given a subset A of $\mathbb{Z}/(N)$, if A contains no non-trivial 3-term arithmetic progression then $|A| = o(N)$

The proof here is in spirit of the graph theoritic approach taken by Conlon, Fox and Zhao.

$\boxed{\text{Proof}}$

Consider $S \subset \mathbb{Z}/(N)$ containing no non-trivial 3-AP. Construct the edge-weighted tripartite graph $G(S) : (X \times Y) \cup (X \times Z) \cup (Y \times Z) \longrightarrow [0, 1]$ where X, Y, Z are all copies of $\mathbb{Z}/(N)$. Set $G(S)|_{X\times Y}(x, y) = \mathbf{1}_S(2x + y)$, $G(S)|_{X\times Z}(x, z) = \mathbf{1}_S(x - z)$ and $G(S)|_{Y\times Z}(y, z) = \mathbf{1}_S(-y - 2z)$. A zero weight between two points implies that no edge lies between them. There were simpler ways to state these edge-weight conditions but instead it is for the sake of coherency in further development of the proof. Pictorially the graph looks as below



Next note that $(x - z) = (-y - 2z) + (x + y + z)$ and $(2x + y) = (-y-2z)+2\cdot(x+y+z)$ which implies $(-y-2z), (x-z), (2x+y)$ are in an arithmetic progression with common difference (x+y+z). Also, $xyz \in X \times Y \times Z$ forms a triangle if and only if all the bipartite sub-parts containing xy, xz, yz agree together. This implies $\mathbf{1}_S(2x + y) \cdot \mathbf{1}_S(x - z) \cdot \mathbf{1}_S(-y - 2z)$ is not zero or $\{2x + y, x - z, -y - 2z\} \subset S$. But $\{x, x + d, x + 2d\} \subset S$ for

$x, d \in \mathbb{Z}/(N)$ should imply $d = 0$. Finally we have $x + y + z = 0$. This implies if xy forms an edge, *a priori* an unique vertex $z = (-x - y)$ exists with edges $xz \in X \times Z$ and $yz \in Y \times Z$. Since every edge in X is joined by some edge in Y agreeing with the linear form aforementioned and each edge forms a triangle with a unique vertex, $G(S)$ has every edge lying in a unique triangle. Thus $G(S)$ has $N^2 = o(N^3)$ triangles. Next by a triangle removal lemma, which states that a graph on n vertices with $o(n^3)$ triangles can be made triangle free by removing $o(n^2)$ edges, $G(S)$ can be made triangle free by removing $o(N^2)$ many edges. Next by previous arguments it follows that $|\{\text{triangles in } G(S)\}| = |\{\text{edges in } G(S)\}|/3$ and since $|S|$ many edges are joined to each vertex (for each fixed choice of a vertex, another vertex is selected agreeing with the linear forms aforementioned from a translation of S in $|S|$ many ways) yielding a total of $3N|S|$ edges, $o(N^2) = |\{\text{triangles in } G(S)\}| = N \cdot |S|$. This finally implies the result.

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Since for practical purposes $(N, 2) = 1$, the graph $G(S)$ may be embedded in $\text{CayleySUM}(\mathbb{Z}/(3N), S)$ in a natural manner (Consider $\pi : e(G(S)) \to e(\text{CayleySUM}(\mathbb{Z}/(3N), S))$ specified by $\pi(xy) := (x(y + N))\mathbf{1}_S(x + y)$ where $\pi(xy) = 0 \implies$ xy is not an edge). Loosely speaking this will be useful because the density of two sets can then be compared by the corresponding density of their respective embeddings in their Cayley-Sum graphs in a natural manner. To be precise if S' is dense in $\mathbb{Z}/(3N)$ then $\text{CayleySUM}(\mathbb{Z}/(3N), S')$ is dense in the complete graph on N vertices since

$$\big(|\{x, y \in \mathbb{Z}/(3N) : \pi(xy) \text{ forms an edge}\}| = 3N|S'| \gtrsim_\delta N^2 = O(3N^2)\big)$$

. Symmetrically an alternative direction may also be proved. Since relative Szemeredi's theorem asserts the existence of arithmetic progressions in a sparse set A (embedded in a sufficiently pseudo-random host set), a dense analog for this sparse set $\widetilde{A}$ will be considered in such a manner that the average of A over some subset S of $\mathbb{Z}/(N)$ will get asymptotically closer to the average of $\widetilde{A}$ over $\mathbb{Z}/(N)$ with N getting larger. Naturally we can form such an $\widetilde{A}$ by drawing $9N|S|$ edges to $\pi(G(A))$ and the resultant graph will be the $\pi(G(\widetilde{A}))$. To be more specific since *a priori* $|\{G(A) \hookrightarrow K_{N,N,N} \text{ is a homomorphism}\}| = o(N^2)$, some edges may be removed from $K_{N,N,N}$ to yield the subgraph $G(S)$ which will be sparse in the sense of number of embeddings in the complete tripartite graph will be sub-quadratic in nature. The commutative diagrams (in two parts) will summarize this:

$$\begin{tikzcd} G(A) \arrow[r, hook, "i_{G(A)}"] \arrow[dr, hook, "i_{G(S)}"'] & G(S) \arrow[d, hook, "i_{G(A)}"] \\ & K_{N,N,N} \end{tikzcd} \qquad \begin{tikzcd} G(A) \arrow[d, hook, "i_{G(A)}"'] \arrow[dr, hook, "i_{G(A)}"] & \\ G(\widetilde{A}) \arrow[r, hook, "i_{G(\widetilde{A})}"'] & K_{N,N,N} \end{tikzcd}$$

The density of $\widetilde{A}$ in $\mathbb{Z}/(N)$ is positive follows from the trivial observation $(|A|+a)/(|S|+a) > |A|/|S|$ for $a > 0$. The dense model acts as a natural extension for a sparse set A in $\mathbb{Z}/(N)$, embedded in a sparse (pseudorandom) host set $\widetilde{A}$, to $\widetilde{A}$ so that the density if preserved or the number of edges to be removed from $G(S)$ to yield $G(A)$ is roughly the same as the number of edges to be removed from $K_{N,N,N}$ to yield $G(\widetilde{A})$. This stems from the idea that the graph regularity lemma fails for sparse subgraphs of a given host graph ($K_{N,N,N}$ for this purpose) without some additional pseudorandomness condition. To be precise if the sparse graph is a subgraph of some sufficiently pseudo-random or random graph, then an analogous regularity lemma can be devised with an additional density constraint. Sets and graphs will be used synonymously in the sense that density of some subset of a host set will be preserved in the analogous graph versions. The rest is to be understood from the relevant context. For the proof of relative Szemeredi's theorem graph theory will be useful (notational difficulty and to state the pseudo-randomness conditions and the sparse triangle counting lemma later on in the proof.

Before discussing the proof of the dense model theorem it is relevant to discuss the notion of a cut norm. The cut norm of a matrix M of order $m \times n$ as defined by Frieze and Kannan is given by

$$\|M\|_{\square} := \sup_{A \times B \in [m] \times [n]} M(A,B) = \sup_{A \times B \in [m] \times [n]} \left(\sum_{i \times j \in A \times B} M_{ij} \right)$$

In a more generic case, a normalization with an extra multiplicative factor of $1/mn$ is considered. One use of this cut norm is used to find the maximum distance between $M(S,T)$ and sum of $\leq 1/\varepsilon^2$ simple rank 1 matrices (in our use-case these will be matrices with a constant submatrix embedded in the zero matrix, with same dimensions as that of the original matrix, in a natural manner. To be more precise a cut matrix $A := \mathrm{CUT}(R, C, d)$ has entries $A_{ij} := d \cdot \mathbf{1}_R(i) \cdot \mathbf{1}_C(j))$ as $S \subset [m]$ and $T \subset [n]$ are varied. If the distance is small relative to the size of the whole matrix, then the matrix may be deemed to have a substantial structured part. Otherwise the matrix may show similarities with that of a random matrix. The following existence lemma guarantees the existence of such a decomposition with an error $o(mn)$

Decomposition Existence Lemma

Suppose M be a real $m \times n$ matrix. Then there exists cut matrices $D^{(1)}, \cdots, D^{(s)}$, where $D^{(t)} := \mathrm{CUT}(R^t, C^t, d_t)$ with entries $D^{(t)}(i,j) := d_t \cdot \mathbf{1}_{R^t}(i) \cdot \mathbf{1}_{C^t}(j)$ for $1 \leq t \leq s \leq 1/\varepsilon^2 \leq mn$, such that if

$$W^{(t)} := M - \left(D^{(1)} + \cdots + D^{(t)}\right)$$

it satisfies that

$$\|W^{(s)}\|_\square \leq \varepsilon\sqrt{mn}$$

Note that if $t = mn$, then $D^{(i)} := M((i+1) \pmod{m+1}, (i+1) \pmod{n+1})$ will suffice. Hence $1/\varepsilon^2 \leq mn$ is assumed.

$\boxed{\text{Proof}}$ It suffices to show $|W^{(s)}(S,T)| \leq \varepsilon\sqrt{mn} \quad \forall\, S, T \subset [m] \times [n]$. Inductively assume there exists cut matrices $D^{(t)}$ for $t < 1/\varepsilon^2$ such that $W = W^{(t)}$ satisfies

$$\|W\|_F^2 \leq (1 - \varepsilon^2 t)$$

The cut norm used here is normalized and hence from cauchy-schwarz it follows that for a $\{0,1\}$-valued matrix M

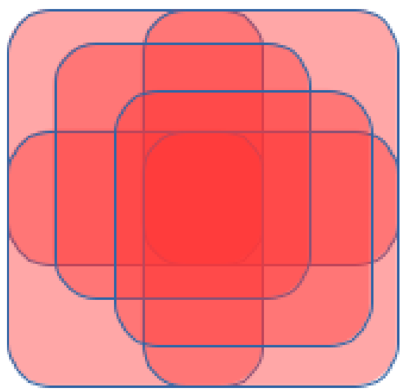
$$\|M\|_\square^{\text{normalized}} \leq \|M\|_F \leq \varepsilon\sqrt{\frac{1}{\varepsilon^2} - t} \leq \varepsilon\sqrt{mn}$$

Henceforth, it suffices to prove the aforementioned inequality. Next, suppose there exists $S' \subset [m]$ and $T' \subset [n]$ such that $|W(S',T')| > \varepsilon\sqrt{mn}$. Set $R^{t+1} := S',\ C^{t+1} := T'$ and $d_{t+1} := |W(S',T')|/|S'||T'|$. Then it follows that,

$$\begin{aligned}
\|W^{(t+1)}\|_F^2 - \|W\|_F^2 &= \|W - D^{(t+1)}\|_F^2 - \|W\|_F^2 \\
&= \sum_{i,j \in R^{t+1} \times C^{t+1}} (2W(i,j) - d_{t+1}) \cdot d_{t+1}
\end{aligned}$$

$$\leq -\varepsilon^2 \|M\|_F^2 \blacksquare$$

From the aforementioned lemma the overlap becomes apparent. It follows that $\|M\|_\square = \|D\|_\square + o(mn)$ where D is the sum of cut-matrices. This suggests that a given matrix can be decomposed into a structured (for which a finite decomposition into cut matrices exists) part and a "random" bounded part. Pictorially the following displays demonstrates the sum of a few cut matrices,



Bounding $\|M\|_\square$ also bounds the maximum overlap possible in the associated structured part, which further bounds the number of decompositions possible. It must be noted that the cut norm is not a function of the size of the matrix: Given a matrix A of order $m \times n$, embed it naturally in a $(m+1) \times (n+1)$ matrix A', such that $A'(i,j) = A(i,j)$ for $1 \leq i \leq m$, $1 \leq j \leq n$, $A'(i, n+1) = -\left(\sum_{1 \leq j \leq n} A(i,j)\right)$ for all $i \in [1,m]$,$A'(m+1, j) = -\left(\sum_{1 \leq i \leq m} A(i,j)\right)$ for all $j \in [1,n]$ and $A'(m+1, n+1) = 0$. It follows that the sum of the entries in A' is zero. This implies summing over any subset containing $m+1$ or $(n+1)$ is the same as summing over their complements

relative to $[1, m+1]$ or $[1, n+1]$. But this sum is similar to a sum with similar index set for A. This implies $\|A\|_\square = \|A'\|_\square$. Loosely speaking the cut norm of a matrix is the measure of it's inherent structure.
In the dense model theorem there is a sparse subset A of $\mathbb{Z}/(N)$ in the sense that $|A| = o(N)$. Naturally, the associated Cayley-Sum graph $\text{CayleySUM}(\mathbb{Z}/(3N), A)$ will be sparse in $K_{N,N,N}$. The associated adjacency matrix of $K_{N,N,N}$ will be the $3N \times 3N$ matrix

$$\begin{bmatrix} \mathbf{0}_N & \mathbf{1}_N & \mathbf{1}_N \\ \mathbf{1}_N & \mathbf{0}_N & \mathbf{1}_N \\ \mathbf{1}_N & \mathbf{1}_N & \mathbf{0}_N \end{bmatrix}$$

Consider A to be sparse in $\mathbb{Z}/(N)$. Since $|\{x, y \in \text{Cay}(\mathbb{Z}/(3N), A) : xy \text{ forms an edge}\}| = 4\times|\{x, y \in G(A) : xy \text{ forms an edge}\}|$, $|e(G(A)| = O(|e(\text{CayleySUM}(\mathbb{Z}/(3N), A)|)$.
Since A is sparse, it follows that $|e(\text{CayleySUM}(\mathbb{Z}/(3N), A)| = o(N^2)$.
A similar argument is possible in the graph theoritic analogue: It follows that

$$\|\text{Adj}(G(A)) - \text{Adj}(K_{N,N,N})\|_\square$$

$$\underset{\|a+b\| \leq \|a\|+\|b\|}{=} \|\text{Adj}(K_{N,N,N}) - \text{Adj}(K_{|A|,|A|,|A|})\|_\square + O(1)$$

where we abuse notation and adjoin $K_{|A|,|A|,|A|}$ with zeros in the rest of the matrix to make the dimension same as the other term. If $A = o(N)$ then the first term becomes $O(N^2)$, which implies a large number of edges are needed to be removed from $K_{N,N,N}$ to yield $K_{N,N,N} \cap \text{Cay}(\mathbb{Z}/(3N), A)$. This implies the cut norm is a useful norm to measure density of a subset in a structured host set (which maybe a group like $\mathbb{Z}/(N)$ in our case).

For the edge-weighted graph

$$\begin{aligned} f : (X \times Y) \cup (X \times Z) \cup (Y \times Z) \to [0, \infty),\ f|_{X\times Y}(x, y) &:= p^{-1}\mathbf{1}_A(2x + y), \\ f|_{X\times Z}(x, z) &:= p^{-1}\mathbf{1}_A(x - z), \\ f|_{Y\times Z}(y, z) &:= p^{-1}\mathbf{1}_A(-y - 2z) \end{aligned}$$

where $p := |A|/N$ is the density of A; $A \subset \mathbb{Z}/(N)$. For the dense case since the choice of A roughly becomes deterministic (similar to how the indicator function of even numbers are useful while the one for the primes are not in the sense there is not sufficient information to give an explicit form with an error which goes away with N growing large, for this sparse subset. Loosely speaking in the dense case there are more structured parts dominating as a total over the rest. This is reminiscent of the decomposition lemma for matrices discussed previously). From previous discussions it is clear that the cayley graphs are natural objects to measure density. We henceforth define the cut norm over $X \times Y$ as

$$\begin{aligned} \|f|_{X\times Y}\|_\square := \sup_{A,B\subset \mathbb{Z}/(N)} |\mathop{\mathbb{E}}_{x,y\in A\times B} f(x,y)| &= \sup_{A,B\subset \mathbb{Z}/(N)} |\mathop{\mathbb{E}}_{x,y} f(x,y)\mathbf{1}_A(x)\mathbf{1}_B(y)| \\ &= \sup_{A,B\subset \mathbb{Z}/(N)} |\mathop{\mathbb{E}}_{x,y} f(2x+y)\mathbf{1}_A(x)\mathbf{1}_B(y)| \\ &= \sup_{A,B\subset \mathbb{Z}/(N)} |\mathop{\mathbb{E}}_{x,y} f(x+y)\mathbf{1}_A(x)\mathbf{1}_B(y)| \end{aligned}$$

where the last line follows since $N \equiv 1 \pmod 2$ and the change of variables $x \leftrightarrow 2x$. Similar for $X \times Z$ and $Y \times Z$. Finally we have $\|f\|_\square := \sup\{\|f|_{X\times Y}\|_\square, \|f|_{X\times Z}\|_\square, \|f_{Y\times Z}\|_\square\}$. A white lie here is that after a change of variables $\|f|_{X\times Y}\|_\square = \|f|_{X\times Z}\|_\square = \|f|_{Y\times Z}\|_\square$. Henceforth for a generic real valued $f : \mathbb{Z}/(N) \to \mathbb{R}$, the cut norm may be defined as

$$\|f\|_\square := \sup_{A,B\subset \mathbb{Z}/(N)} |\mathbb{E}_{x,y} f(x+y)\mathbf{1}_A(x)\mathbf{1}_B(y)|$$

We may define the dense model theorem as follows

Dense Model Theorem

Given $\varepsilon > 0$, there exists a $\delta > 0$ such that the following holds. Let $f, v : \mathbb{Z}/(N) \to [0,\infty)$ such that

$$\|v - 1\|_\square \leq \delta$$

where $f(x) \leq v(x) \quad \forall x \in \mathbb{Z}/(N)$. Then there exists a function $g : \mathbb{Z}/(N) \to [0,1]$ such that

$$\|f - g\|_\square \leq \varepsilon$$

Note that *a priori* $\|f-g\|_\square = o(1) \implies \mathbb{E}f \approx \mathbb{E}g$. Also if $\|v\|_\infty < \infty$, then $g := \mathbb{E}f$ suffices and the aforementioned theorem need not be summoned. Hence f, v will be considered unbounded. The main idea of this theorem is to show f can be decomposed into $g_1 + g_2$ where one part is bounded and the other part is unbounded with small cut norm. This is reminiscent of matrix decomposition. Since this deals with generic real valued functions over finite abelian groups, this decomposition theorem is a generalization of the matrix decomposition theorem but in a more abstract sense.

$\boxed{\text{Proof}}$

Consider $0 < \varepsilon < 1/2$. Next it suffices to find a $g : \mathbb{Z}/(N) \to [0, 1+\varepsilon/2]$ such that $\|f - g\|_\Box \leq \varepsilon/2$ (since it follows that $\|g - g/\left(1+\frac{\varepsilon}{2}\right)\|_\Box \leq \varepsilon/2 \implies \|f - g/\left(1+\frac{\varepsilon}{2}\right)\|_\Box \leq \varepsilon$ and $\widetilde{g} := g/\left(1+\frac{\varepsilon}{2}\right)$ will suffice. Alternatively, $\widetilde{g} := \min\{1, g\}$ would also suffice).

Set $K_1 := \{f : \mathbb{Z}/(N) \to [0, 1+\varepsilon/2]\}$ and $K_2 := \{f : \mathbb{Z}/(N) \to [0, \infty) : \|f\|_\Box \leq \varepsilon/2\}$. It suffices to show $f = g_1 + g_2$ for $g_i \in K_i, i \in \{1, 2\}$. On the contrary, if it does not happen then there exists a function $\psi : \mathbb{Z}/(N) \to \mathbb{R}$ (by Hahn-Banach) such that

(a) $\langle f, \psi\rangle > 1$, and

(b) $\langle (g_1+g_2), \psi\rangle \leq 1 \quad \forall g_i \in K_i$ Define the dual of $\|\cdot\|_\Box$ on the space of all real valued functions with domain $\mathbb{Z}/(N)$ as $\|\psi\|_\Box^* := \sup\limits_{\|f\|_\Box \leq 1} \langle f, \psi\rangle$. Set $f := \mathbf{1}(x)$ as the function supported on just one element. Then $\|f\|_\infty \leq \|f\|_\Box^*$ and on setting $g_1 = 0$, $\langle g_2, \psi\rangle \leq 1 \ \forall \|g_2\|_\Box \leq \varepsilon/2 \underset{(normalization)}{\implies} \|\psi\|_\Box^* \leq 2/\varepsilon$. This finally implies $\|\psi\|_\infty \leq \|\psi\|_\Box^* \leq 2/\varepsilon$. It also follows that $\langle 1, \psi_+\rangle \leq 1/\left(1+\frac{\varepsilon}{2}\right)$ after setting $g_1 := \left(1+\frac{\varepsilon}{2}\right)\mathbf{1}_{\psi>0}, \quad g_2 := 0$.

Next using a polynomial approximation theorem, there exists a polynomial $P := a_0 + xa_1 + \cdots + x^d a_d$ which approximates $\max\{0, \psi\}$ such that

$$\|P\psi - \psi_{>0}\| \leq \varepsilon/20$$

It follows that

$$\|P\psi\|_\Box^* \leq \sum_{0\leq i\leq d} |p_i|\|\psi^i\|_\Box^* \leq \sum_{0\leq i\leq d} |p_i|(\|\psi\|_\Box^*)^i \leq \sum_{0\leq i\leq d} |p_i|\left(\frac{2}{\varepsilon}\right)^i =: R(\text{say})$$

Next, set

$$\delta := \min\{1, \varepsilon/(2R)\}$$

A normalization argument implies

$$|\langle v-1, P\psi\rangle| \leq \|v-1\|_\Box \|P\psi\|_\Box^* \leq \varepsilon/2$$

where the last inequality follows from the assumption at the beginning.
It finally follows that,

$$\begin{aligned}\langle v, P\psi\rangle &= \langle 1, P\psi\rangle + \langle v-1, P\psi\rangle \\ &\leq \langle 1, \psi_+\rangle + \varepsilon/20 + \varepsilon/20 \\ &\leq \frac{1}{1+\frac{\varepsilon}{2}} + \varepsilon/10\end{aligned}$$

where the first inequality followed from the polynomial approximation step and the bound found recently.
It follows from a trivial bound that

$$\langle v, \psi_+ - P\psi\rangle \leq \|v\|_1 \|\psi_+ - P\psi\|_\infty \leq \varepsilon/10$$

Finally since $0 \leq f \leq v$,

$$\langle f, \psi\rangle \leq \langle v, \psi_+\rangle \leq \langle v, P\psi\rangle + \langle v, \psi_+ - P\psi\rangle \leq \frac{1}{1+\frac{\varepsilon}{2}} + \varepsilon/5 \leq 1$$

The last expression is a contra-supposition. ■

The concerned dense model $\widetilde{A}$ may be formed by constructing the set where $x \in \mathbb{Z}/(N)$ is included with probability $g(x)$. A more involved dense model theorem as in the original paper of Green and Tao uses the Gower's norms which is currently out of scope of understanding of the writer.
The proof of relative Roth's theorem is decomposed into two parts
(A) To find a dense model $\widetilde{A}$ of the sparse set A
(B) Using the 3-AP count in $\widetilde{A}$ to find the 3-AP count in A using a sparse triangle counting lemma

It remains to show the second part. This mainly concerns the edge-weighted graph constructed before as follows:

$$f : (X \times Y) \cup (X \times Z) \cup (Y \times Z) \to \mathbb{R},\ \begin{aligned} f|_{X\times Y}(x,y) &:= p^{-1}\mathbf{1}_A(2x+y), \\ f|_{X\times Z}(x,z) &:= p^{-1}\mathbf{1}_A(x-z) \quad, \\ f|_{Y\times Z}(y,z) &:= p^{-1}\mathbf{1}_A(-y-2z)\end{aligned}$$

Note that in a more abstract setting, we construct the edge-weighted bipartite graphs, $g_{XY} : X \times Y \to \mathbb{R}, g_{XZ} : X \times Z \to \mathbb{R}$ and $g_{YZ} : Y \times Z \to \mathbb{R}$ where the edge weights will be specified by an abstract function (which will be the normalized edge-weighted indicator function in practical purposes) $g : \langle X, Y, Z \rangle \to \mathbb{R},$ where it is defined on the group generated by sets X, Y, Z and $g_{XY}(x, y) := g(2x + y),\ g_{XZ} := g(x - z)$ and $g_{YZ} := g(-y - 2z)$

The rest of the proof is used to devise a triangle counting lemma in the sparse setting. The final proof becomes apparent after that. The author does not have anything more to state in a noble manner.